Diploma Verification Through Multi-Modal Image Processing: Face Detection, Perforation Text Recognition, and System Architecture Evaluation

1st Muh. Iqbal Ahmad
Departement of Informatics
Hasanuddin University
Makassar, Indonesia
ahmadmi20d@student.unhas.ac.id

2nd Ingrid Nurtanio
Departement of Informatics
Hasanuddin University
Makassar, Indonesia
ingrid@unhas.ac.id

3rd Zahir Zainuddin
Departement of Informatics
Hasanuddin University
Makassar, Indonesia
zahir@unhas.ac.id

2024 7th International Conference on Information and Communications Technology (ICOIACT) | 979-8-3315-3620-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICOIACT64819.2024.10913411

***Abstract*—This research presents a novel approach to digitally verify diploma authenticity using a combination of Siamese Neural Network (SNN) for face verification and YOLOv8 for text perforation detection. The SNN model, utilizing a binary cross-entropy loss function, successfully minimized the loss value from 0.169 to 0.0025 by the sixtieth epoch, achieving an accuracy of 0.98. This demonstrates the model's high proficiency in differentiating between anchor and positive/negative images. Additionally, the YOLOv8 model achieved a mean Average Precision (mAP) of 0.93 at a 0.5 IoU threshold, further enhancing the precision of text perforation detection. This integrated system offers a robust and efficient method for ensuring the authenticity of diploma documents, contributing significantly to digital document security and forgery prevention. The research supports the implementation of deep learning in official document verification.**

Keywords—diploma document, face verification, perforations detection, machine learning, deep learning.

I. INTRODUCTION

In the modern era, many people pursue instant gratification unethically, disregarding the consequences. This drive for self-satisfaction, often disproportionate to one's efforts, leads individuals to employ various means to achieve their goals. Higher education, for instance, is essential for many, yet its complexities involve not only academic performance but also oversight by institutions and the government. Technological advances, such as digital image processing tools like Adobe Photoshop, have facilitated forgery, especially in education, where certificates and diplomas are falsified for personal gain at the expense of others[1]

A diploma is a vital document from universities, proving that an individual has completed their education and passed the necessary exams. Traditional diplomas, being physical, are prone to loss or damage, and verifying their authenticity is difficult without contacting the issuing authority. The current process requires direct verification and legalization through the issuer's signature. Diplomas from accredited institutions are crucial for employment, and due to high competition in Indonesia, cases of fake diplomas are common, often reported in the media[2]

Software like Adobe Photoshop and GIMP provides various tools for image manipulation, including editing. Due to this, a method is needed to help the public verify the authenticity of digital images. Diploma forgery has become a notable issue, as fake diplomas are often used for job applications or to gain official positions. This practice is not limited to ordinary individuals but also includes public figures and officials who buy degrees to gain respect or appear more intelligent[2]

- Face Detection: The system detects and locates faces within an image or video stream by scanning the input data to identify regions of interest (ROI) corresponding to human faces. This process is carried out using advanced algorithms, often based on cascading classifiers or deep convolutional neural networks (CNN). As demonstrated in [3], a CNN cascade for face detection was developed with a strong discriminative ability, leading to highly effective performance.
- Feature Extraction: After detecting faces, the next vital step is extracting unique facial features. This process converts the facial image into a set of numerical representations called feature vectors. Deep learning-based feature extraction using Convolutional Neural Networks (CNNs).
- Verification or Recognition: The final stage involves matching extracted facial feature vectors with a database using methods advanced Deep Learning techniques are also employed for identity verification. The system assigns an identity label based on the closest match, determined by the chosen distance metric or classification method.

Research on perforation in security documents remains limited, with manual inspection methods being time-consuming and error-prone. Few efforts focus on automated systems combining local (hole shape) and global (alignment) feature analysis, while the lack of public datasets and reliance on basic image processing hinder progress. This highlights the need for advanced, efficient solutions to improve forgery detection[4].

The model's dependence on prior high-quality images of original artworks limits its use when such images are unavailable. Integrating traditional AI techniques could improve its accuracy and adaptability for various artwork and counterfeit detection tasks. Future research could explore methods to authenticate artworks without prior image data, expanding the model's applicability [5].

The current study reveals that existing face detection algorithms struggle with identity documents under complex

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:00:41 UTC from IEEE Xplore. Restrictions apply.

real-world conditions. Future research should focus on developing specialized algorithms or enhancing existing ones to improve robustness and accuracy in face detection across diverse and challenging scenarios, particularly for documents with varying image qualities and environmental conditions [6].

The study shows that trained forgery detectors surpass humans in identifying manipulated images, but more adaptable and generalizable methods are needed. Current models may struggle with novel or uncommon manipulation techniques not covered in training data. Future research should focus on algorithms that quickly adapt to new forgeries with minimal training, improving robustness in real-world scenarios. Additionally, exploring the ethical and legal implications of synthetic media and establishing detection standards are critical areas for further study. [7].

While batch correction improves efficiency and accuracy, further research should explore active learning techniques to better identify and prioritize error clusters for post-processing. Future work could also focus on adapting the OCR recognizer itself to handle document collections, reducing error rates. Dynamic updates to the OCR model based on document characteristics could enhance system robustness and performance. [8].

While the hybrid approach helps overcome limited annotated data for OCR training, more research is needed to improve the quality and diversity of synthetic data, particularly for historical documents. Advancing character generation and line synthesis techniques that capture historical script nuances could boost accuracy. Additionally, integrating diverse historical datasets, including handwritten and degraded documents, would enhance the OCR system's robustness across various types of texts. [9].

Research on perforation in forensic document examination lacks automation, relying heavily on manual methods that are time-consuming and error-prone. The absence of advanced computational approaches highlights the need for automated solutions to improve accuracy and efficiency in analyzing perforation characteristics[10].

II. Proposed Method

This study presents two methods for verifying diploma authenticity. The first method uses face-based verification, where the Haar cascade algorithm detects facial images from diploma documents from the different levels in Table 1. Features are extracted using a Siamese Neural Network (SNN) and compared with official student photos from prior academic records, such as junior and senior high school images. Binary cross entropy loss is then used to evaluate the accuracy of the matches. The second method focuses on detecting and validating perforated text, typically located in the center-right section of the diploma, which includes key details like the year of issuance and education level (e.g., "D-2020" for primary education, "M-2020" for secondary education). The YOLO object detection model identifies the perforation. The results are combined to enhance the overall accuracy and reliability of the diploma verification system.

A. Scenario 1

In the first scenario, we use face-based verification by applying the Haar cascade algorithm for its accuracy and efficiency in detecting faces. A pre-trained Siamese Neural Network (SNN) architecture is employed for feature extraction, which captures hierarchical facial details. We then compare the results using SNN and Binary cross-entropy loss. SNN helps detect intricate facial patterns, providing a stronger basis for comparison, ultimately improving the accuracy of the face recognition model.

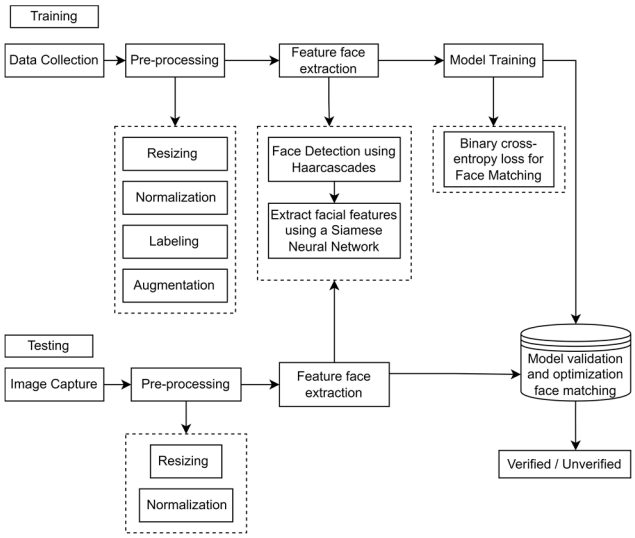


Fig 1. Block diagram of face variable

The following outlines each step in the facial verification process, which uses a Siamese Neural Network (SNN) for both the training and testing phases.

1. *Data Collection*

The initial phase of training involves gathering a substantial dataset of facial images, primarily sourced from diploma documents. To enhance the model's efficiency and generalization, this data is expanded through augmentation techniques, which aim to create a more robust and diverse training set.

2. *Pre-Processing*

Once the data is collected, it undergoes further processing to prepare it for the model. This stage involves several key sub-processes, including:

a) Resizing: Adjusting the image dimensions to align with the model's input size (typically a fixed size, such as 250x250 pixels for SNN).
b) Normalization: Converting each pixel value to a standardized range (e.g., 0 to 1) to accelerate convergence during training.
c) Labeling: Assigning labels to the images, indicating whether two images depict the same person or different individuals.
d) Augmentation: Applying various transformations to the face images, such as rotation, flipping, or changes in lighting, increases the dataset's diversity and enhances the model's generalization ability.

3. *Feature Face Extraction*

Following pre-processing, the Haarcascades algorithm is applied to each diploma document image to

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:00:41 UTC from IEEE Xplore. Restrictions apply.

detect faces. The identified faces are then processed into feature representations using a Siamese Neural Network, which compares the embeddings or feature vectors of three faces for verification.

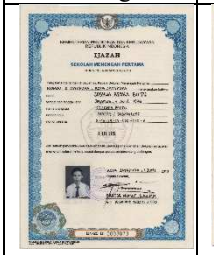
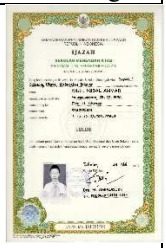
4. Model Training

At this stage, the Siamese Neural Network (SNN) is trained for face matching by generating embedding vectors for pairs of face images. The network learns to differentiate between matching and non-matching faces by minimizing a binary cross-entropy loss or triplet loss. The model adjusts its parameters to ensure that similar embeddings represent faces from the same person while distinct embeddings represent faces from different people.

5. Model Validation and Optimization

Once the Siamese Neural Network (SNN) has been trained, the next step is validation and optimization. The model is evaluated using validation data to assess its accuracy in verifying face matching. Adjustments are made to the threshold for similarity and hyperparameters to improve the model's performance in distinguishing between matching and non-matching faces. This ensures that the system reliably identifies faces with high precision and performs consistently in face verification tasks.

TABLE I. DIPLOMA DOCUMENTS FROM DIFFERENT LEVELS

Educational Level		
Junior High	Senior High	Senior High
		

a) SNN (Siamese Neural Network)

In a Siamese Neural Network (SNN), two facial images pass through identical networks with shared weights, generating feature embeddings (Features 1 and 2) that capture key facial characteristics, as shown in Figure 2. A classifier then compares the features using a distance metric, like Euclidean distance, to assess similarity. If the distance is below a set threshold, the faces are classified as the same person; otherwise, they are different. This architecture excels in facial verification by effectively learning and comparing image similarities.

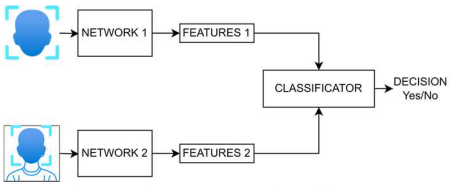


Fig 2. Siamese network architecture

B. Scenario 2

In the second method, the system detects and verifies perforated text in the center-right of the diploma, where crucial details like the year and education level are marked. A YOLO object detection model identifies the perforation and validates its authenticity, working in tandem to increase the accuracy and reliability of the diploma verification process.

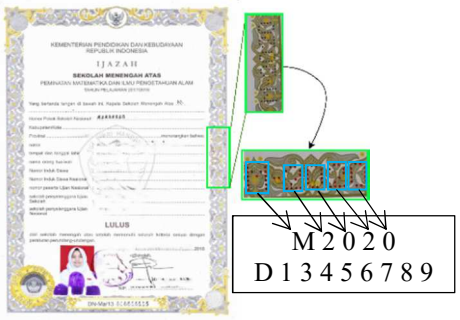


Fig 3. The perforated text images and text classes

Figure 3 shows perforated text images, a key anti-counterfeiting method in security documents. These images consist of small holes created through needle or laser perforation with diameters between 0.1 and 0.8 mm[4].

1) Perforations Detection

This research aims to develop a system capable of detecting perforations in scanned diploma documents using computer vision techniques, specifically by applying the "You Only Look Once" (YOLO) algorithm.[11] YOLO, initially created by Joseph Redmon, a PhD candidate at the University of Washington, has evolved through several iterations, including YOLOv2 and YOLOv3.

The primary goal of instance segmentation is to identify and classify each distinct object within an image, such as various types of text perforations, by labeling each pixel associated with that object. This method allows for the differentiation of different perforation types by generating an image where every pixel is annotated, revealing the object it corresponds to, and specifying the associated perforation text.

a) *Preprocessing Stage*

In the preprocessing stage, images are gathered from various sections of the object, annotated, and then utilized to train the YOLOv8 model using the free GPU available on Google Colab. The model's effectiveness is assessed through performance metrics such as Precision, Recall, and Mean Average Precision (mAP).

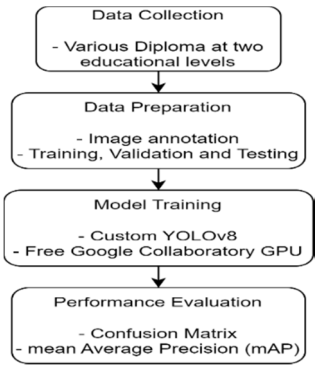


Fig 4. Block Diagram of Perforation Year Detection

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:00:41 UTC from IEEE Xplore. Restrictions apply.

1) Data Collection

The diploma images were captured using both a scanner and a mobile phone, resulting in varying resolutions. Subsequently, these images were resized to 640 x 640 pixels to improve processing speed.

2) Data Preparation

During this phase, subject identification and detection in the images are performed using polygon masks for annotation. Once the annotation is completed, the dataset is divided into three subsets: training, validation, and test sets. Specifically, 90% of the data is allocated for training, 6% for validation, and 4% for testing, with this division being handled through the Roboflow platform.[12]. The dataset is categorized into 13 text classes: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, D, M and Perforation, as shown in Fig 3.

3) Training Model

The YOLOv8 model is trained using the free Graphics Processing Unit (GPU) available through Google's collaborative platforms. This training consists of several cycles, known as epochs, with the entire dataset being processed over 137 epochs.

4) Performance Evaluation

At this stage, the effectiveness of the model is evaluated. Using advanced deep learning techniques, the evaluation includes metrics such as Precision, Recall, accuracy rate, confusion matrix, and Mean Average Precision (mAP).

a) Training Stage

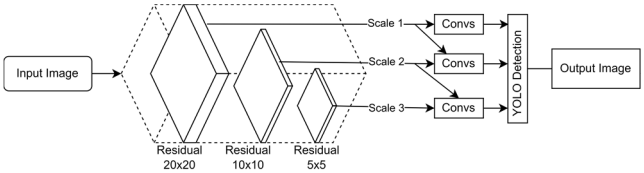


Fig 5. YOLOv8 Architecture [11]

The YOLO (You Only Look Once) algorithm was initially proposed by Joseph Redmon, a computer vision researcher [11]. It processes diploma images at a resolution of 640x640 pixels. At its foundation, a deep learning neural network leverages convolutional layers within the Darknet framework [13]. To extract features. The algorithm segments the image into a grid, with each segment responsible for predicting objects within its region.

1) Model Evaluation

Mean Average Precision (mAP) is an essential metric used to evaluate the effectiveness of object detection models across various classes. It is calculated by determining the average precision for each category. It is derived by evaluating the area under the precision-recall curve and is analyzed using the Intersection over Union (IoU) metric. Once the average precision for each class is determined, the mean of these values is computed to yield the overall performance score.

$$mAP = \frac{1}{N}\sum APi \qquad (1)$$

In the given equation, N(1) represents the total number of classes, while AP denotes the average precision for each class. Therefore, mAP(1) evaluates a system's overall performance in accurately identifying objects across various classes by averaging the precision values for each class. This metric provides a comprehensive assessment of the system's ability to detect and classify objects correctly across the entire dataset. (1)

2) Precision and Recall

$$Precision = \frac{TP}{TP + FP} \qquad (2)$$

$$Recall = \frac{TP}{TP + FN} \qquad (3)$$

Precision is the ratio of True Positives (TP) to the sum of True Positives and False Positives (FP), calculated as Precision = TP / (TP + FP). It reflects how many of the model's positive predictions are correct. Recall, or Sensitivity, measures the model's ability to identify all actual positives, calculated as Recall = TP / (TP + FN). It evaluates how well the model detects true positive cases.

3) Confusion Matrix

A confusion matrix is used to assess the performance model. In binary classification, it includes True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), representing the model's correct and incorrect predictions. Key metrics like accuracy, precision, and recall are derived from it. For multi-class classification, the matrix expands to an n x n table, where n is the number of classes. This is crucial in cases where certain errors have more significant consequences than others.[14].

III. Result And Discussion

In this section of results and discussion, we study the complexity of diploma verification in two different scenarios, evaluating their performance in recognition and accuracy across datasets consisting of 200 to 500 individuals. Our experiments reveal interesting patterns on how these recognition methods function across different dataset sizes using our proposed model, which provides valuable insights into our research.

a) Dataset Description

We used a scanner and a cell phone to capture diploma documents in detail. This effort resulted in our primary facial images on diploma dataset, which includes 20-30 sample images for each of the approximately 200 students. In addition, we increased the diversity and completeness of the dataset by integrating the LFW dataset, selecting 360 people for inclusion.



(a) (b)

Fig 6. Data collection: (a) primary face dataset, (b) secondary face dataset

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:00:41 UTC from IEEE Xplore. Restrictions apply.

We curated a dataset of approximately 2,000 grayscale images representing 200 individuals (Figure 6a), sourced from diploma documents with faces automatically cropped using OpenCV. To enhance this, we included a subset of the Labeled Faces in the Wild (LFW) dataset (Figure 6b), focusing on 200 individuals with at least three samples for comparison.

b) Scenario 1 (Face-Based Verification)

In our proposed system, we assess two different scenario models, the first scenario model is outlined in Table 2. In Table 2, the initial step involves face detection from the input image, followed by preprocessing tasks such as grayscaling and cropping using Haaarcascade. Next, the system proceeds to recognize the input image for person identification. It then presents a feature approach with three stages of detection: first detecting anchors and positives, and then detecting negative face samples.

TABLE II. FACE VARIABLE RECOGNITION PROCESS

Detection	Preprocessing & Feature Extraction			Recognition
	Positive	Anchor	Negative	
				

Triplet Loss in a Siamese Neural Network minimizes the distance between similar face embeddings (Anchor and Positive) while maximizing the distance between dissimilar ones (Anchor and Negative). The network processes three inputs, Anchor, Positive, and Negative, through shared weights to generate feature embeddings. The Anchor is the reference image, the Positive is from the same person, and the Negative is from a different one. The objective is to make the Anchor-Positive distance smaller than the Anchor-Negative distance by a set margin, enhancing face recognition accuracy.

TABLE III. MEAN PREDICTION MODEL SNN

Epoch	Loss	Recall	Precision
10	0.169	0.591	0.878
20	0.066	0.882	0.968
30	0.083	0.943	0.990
40	0.467	0.939	0.984
50	0.0270	0.975	0.990
60	0.0025	0.986	0.991

A binary cross-entropy loss function was used to evaluate the model's ability to determine if two input images (anchor and positive/negative) belonged to the same person. In Table 3, the loss dropped significantly from 0.169 in the first epoch to 0.0025 by the sixtieth, showing the model's learning efficiency. Accuracy also improved from 0.591 in the first epoch to over 0.975 by the fiftieth, reaching 0.986 by the sixtieth, indicating the model's strong performance in distinguishing between the image pairs.

c) Scenario 2 (Perforation Detection)

The model was trained on 200 diploma document images over 137 epochs, achieving a mean Average Precision (mAP) of 0.935 at 50% detection threshold (mAP50) and 0.761 across 50%-95% thresholds (mAP50-95). The optimal number of epochs was determined by tracking the steady improvement in mAP scores during training. Table 4 shows the comparison of mAP50 and mAP50-95 across epochs, highlighting the model's consistent learning progress, with peak performance at the 137th epoch.

TABLE IV. COMPARISON OF MAP 50 WITH MAP 50-95 ACROSS EACH EPOCH AND EPOCH STEPS

Epoch	mAP50	mAP50-95
10	0.339	0.195
30	0.650	0.492
50	0.822	0.612
70	0.867	0.672
90	0.893	0.733
127	0.934	0.765
137	0.935	0.761

Furthermore, the precision and recall results highlight the model's performance, which was trained over 137 epochs on various diploma document classes, such as Perforation, D,M,0,1,2,3,4,5,6,7,8 and 9. Fig 7 illustrates the mAP50-95 throughout the training process for each class at different epochs.

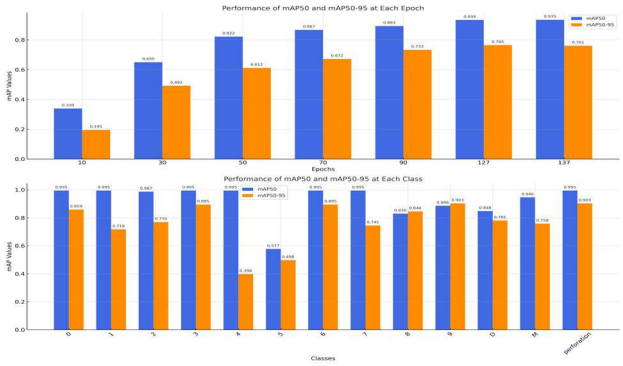


Fig 7. mAP50-95 at each epoch and class bar chart

The training data includes 200 diploma document images with orientation variations, both normal and tilted. The results, shown in the accompanying illustration in Fig 8, demonstrate the model's ability to classify diploma documents such as perforation, D,M,0,1,2,3,4,5,6,7,8, and 9. The illustration highlights the recognition outcomes for these document levels.

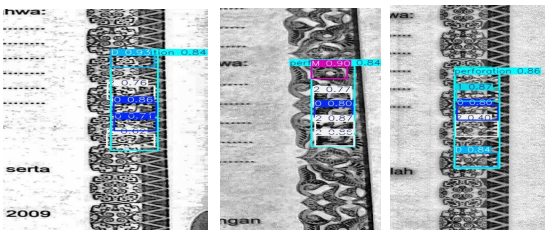


Fig 8. Recognition results perforation.

d) Performance Evaluation

The training dataset consisted of 300 diploma documents, including 630 labeled face images (anchor, positive, negative) and text perforation detection, classified as "verified" or "unverified." Image augmentation was applied to improve quality. A validation process followed to assess the model's generalization to unseen data, evaluating two scenarios using Mean Average Precision (mAP) and training loss. The results are summarized in Table 5.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:00:41 UTC from IEEE Xplore. Restrictions apply.

TABLE V. TESTING OF THE MODEL

model result	Overall model results	
	val_map_05	train_loss
SNN	0.98	0.0025
Perfo_YOLO V8	0.93	2.55

The table compares two models: Siamese Neural Network (SNN) for facial verification and YOLOv8 for text perforation detection. The SNN achieved a "val_map_05" of 0.98 with a low training loss of 0.0025, showing strong optimization in distinguishing face pairs. YOLOv8 outperformed in text detection with a "val_map_05" of 0.93, though it had a higher training loss of 2.55, typical for complex object detection tasks. Both models excelled, with YOLOv8 leading in text detection precision and SNN in training optimization for facial verification.

TABLE VI. TESTING SCENARIO MODEL

No	Test Data	Face-Based Verification	Perforation Detection	Validation
1	Test Data 1	Verified	Verified	Valid
2	Test Data 2	Verified	Verified	Valid
3	Test Data 3	Verified	Verified	Valid
4	Test Data 4	Unverified	Unverified	Invalid
5	Test Data 5	Verified	Verified	Valid
6	Test Data 6	Unverified	Unverified	Invalid
…	…	…	…	…
96	Test Data 96	Verified	Verified	Valid

In this testing scenario presented in Table 6, the model evaluates 96 test data. The model's predictions are compared with the actual values. The “verified” class is represented by the value 1, which shows that diploma documents were successfully detected for the same person and perforation. The value 0 represents the “unverified” class, indicating that diploma documents did not detect the same person and perforation. The Confusion Matrix table can be seen in Table 7.

TABLE VII. THE CONFUSION MATRIX

		Actual Value	
		Verified	Unverified
Model Prediction	Verified	46	2
	Unverified	1	47

IV. CONCLUSION

This study demonstrates that the Siamese Neural Network (SNN) method for face verification when combined with text perforation detection on documents, delivers highly accurate results in digitally verifying the authenticity of diploma documents. Using a binary cross-entropy loss function, the SNN model significantly reduced the loss value from 0.169 in the initial epoch to 0.0025 by the sixtieth epoch, showcasing its ability to learn patterns and minimize prediction errors. The model’s accuracy consistently improved during training, reaching a peak value of 0.98 in several epochs, reflecting its effectiveness in distinguishing between anchor images and positive/negative pairs. Additionally, text perforation detection on diploma documents, using the YOLOv8 model, achieved a mAP of 0.5 at 0.93, further validating its performance. The combination of multi-level face matching and visual feature detection provides a robust and efficient solution for verifying document authenticity. This approach is crucial in enhancing the security of digital documents and preventing forgery and can be adapted for various official document verification applications. The findings of this research support the use of deep learning methods in document verification and reveal significant potential for further advancements in the future.

REFERENCES

[1] E. Ardhianto and N. Wakhidah, “Pengembangan Metode Otentikasi Keaslian Ijasah Dengan Memanfaatkan Gambar Qr Code,” *Jurnal Transformatika*, vol. 13, no. 2, p. 35, 2016, doi: 10.26623/transformatika.v13i2.325.

[2] W. R. Putri, A. W. Widodo, and B. Rahayudi, “Penerapan Ciri Geometric pada Deteksi dan Verifikasi Tanda Tangan Offline,” *Jurnal Pengembangan Teknologi Informasi dan Ilmu Komputer (JPTIIK) Universitas Brawijaya*, vol. 2, no. 5, pp. 1960–1969, 2018, [Online]. Available: https://docplayer.info/53150196-Penerapan-ciri-geometric-pada-deteksi-dan-verifikasi-tanda-tangan-offline.html

[3] H. Li, Z. Lin, X. Shen, J. Brandt, and G. Hua, “A Convolutional Neural Network Cascade for Face Detection.”

[4] E. Zhang and Y. Zhang, “Forgery detection for perforated number in security document by analyzing the perforated holes,” *Imaging Science Journal*, vol. 65, no. 1, pp. 40–48, Jan. 2017, doi: 10.1080/13682199.2016.1257674.

[5] M. Mirbod, “Artwork’s Authenticity Recognition Model Using Machine Vision and Applications,” 2023, doi: 10.21203/rs.3.rs-3758505/v1.

[6] S. Bakkali, M. M. Luqman, Z. Ming, and J.-C. Burie, “Face Detection in Camera Captured Images of Identity Documents Under Challenging Conditions,” Institute of Electrical and Electronics Engineers (IEEE), Nov. 2019, pp. 55–60. doi: 10.1109/icdarw.2019.30065.

[7] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Niessner, “FaceForensics++: Learning to detect manipulated facial images,” in *Proceedings of the IEEE International Conference on Computer Vision*, Institute of Electrical and Electronics Engineers Inc., Oct. 2019, pp. 1–11. doi: 10.1109/ICCV.2019.00009.

[8] D. Das, J. Philip, M. Mathew, and C. V. Jawahar, “A cost-efficient approach to correct OCR errors in large document collections,” in *Proceedings of the International Conference on Document Analysis and Recognition, ICDAR*, IEEE Computer Society, Sep. 2019, pp. 655–662. doi: 10.1109/ICDAR.2019.00110.

[9] J. Martinek, L. Lenc, P. Kral, A. Nicolaou, and V. Christlein, “Hybrid training data for historical text OCR,” in *Proceedings of the International Conference on Document Analysis and Recognition, ICDAR*, IEEE Computer Society, Sep. 2019, pp. 565–570. doi: 10.1109/ICDAR.2019.00096.

[10] J. Seaman Kelly and M. Angel, “Forensic Document Examination in the 21st Century,” 2021.

[11] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, IEEE Computer Society, Dec. 2016, pp. 779–788. doi: 10.1109/CVPR.2016.91.

[12] F. Ciaglia, F. S. Zuppichini, P. Guerrie, M. McQuade, and J. Solawetz, “Roboflow 100: A Rich, Multi-Domain Object Detection Benchmark,” Nov. 2022, [Online]. Available: http://arxiv.org/abs/2211.13523

[13] T. Diwan, G. Anirudh, and J. V. Tembhurne, “Object detection using YOLO: challenges, architectural successors, datasets and applications,” *Multimed Tools Appl*, vol. 82, no. 6, pp. 9243–9275, Mar. 2023, doi: 10.1007/s11042-022-13644-y.

[14] I. Markoulidakis, I. Rallis, I. Georgoulas, G. Kopsiaftis, A. Doulamis, and N. Doulamis, “Multiclass Confusion Matrix Reduction Method and Its Application on Net Promoter Score Classification Problem,” *Technologies (Basel)*, vol. 9, no. 4, Dec. 2021, doi: 10.3390/technologies9040081.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 09,2026 at 07:00:41 UTC from IEEE Xplore. Restrictions apply.